to be segmented for low mobility and Fixed Wireless Access (FWA)

These bands have also been grouped into the release tiers mentioned earlier, with the Study Tier of 3600-3700 MHz becoming a shared band internationally. It will require significant technological advancement for real-time coordination and has been recommended for a rollout only after such technology is available. Other tiers have bands recommended for free usage for two years to promote trials and indigenous R&D. This is necessary to encourage various service providers to deploy their 5G networks in India, and soften the blow of initial infrastructure investment that is required to move onto the next generation of cellular connectivity.

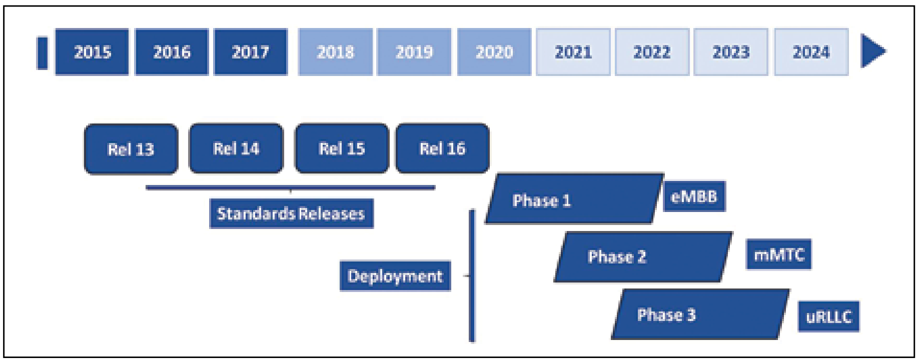
### Radio Backhaul Spectrum

The radio backhaul connects base stations to backend networks and the internet cloud, and require a high density of operations, up to 100-200 per Sq. km. Due to the high density, it favours wireless technologies like Millimetric Band Distribution Networks. This will need a wideband spectrum. As a part of this report, the usage of 57-71 GHz as unlicensed and 71-76 GHz and 81-86 GHz under a light touch licensing regime has been recommended for backhaul and access links.

### 5 GHz WiFi Spectrum

Currently, India has only the limited 50MHz spectrum available for an unlicensed band for outdoor WiFi. According to the committee, this needs to be expanded to 5150 - 5350 MHz and 5725 - 5825 MHz bands as unlicensed for outdoor use, and the 5470-5725 MHz band as unlicensed for indoor / outdoor use maintaining DFS compliance. The Indian telecom regulator has recommended auctioning 20MHz blocks in the 3,300-3,600 MHz band for 5G services at a price of ₹492 crores per MHz. In South Korea, the same band was priced at much less in auctions held in June.

Apart from this, the report also recommends setting up a Standing Committee with a 5-year-term to advise the Government on building up India's Spectrum technology capability, as well as build spectrum management skills and infrastructure in India to deal with the growing complexity of spectrum use in India, to bring it up to international standards. The report also suggests that the government should try to bring international 5G conferences to India as well as set up national 5G events, and the creation of a comprehensive programme to develop India-specific 5G applications. A broad budgetary estimate of ₹300 crores in the first year, ₹400 crores in the second, ₹500 crores in the third and ₹400 crores in the fourth year has been announced. It also goes on to specify how 5G trials should be conducted, with involvement from BSNL, Jio, Airtel and Vodafone as well as provision of financial support. With these plans in place, we need to look at how the carriers are responding.



The recommended timeline of future releases for India

## WHAT ARE THE PROVIDERS DOING?

Before we try and understand what different network providers in India are doing for 5G deployment, we need to understand the difference between a 5G-ready network and a 5G network. The former indicates that a provider has the necessary infrastructure in place to roll out 5G when the spectrum allocation has been completed, without significant additional investments required, whereas the latter implies a fully functional, demonstrated 5G network that satisfies the standard itself.

In August 2018, TRAI had set the pricing for the spectrum at ₹492 crores per unit as the starting price for auctioning 5G airwaves in the 3300-3600 MHz bands. A recently concluded 5G spectrum audition in Korea, comparatively, was at ₹65 crore per unit, according to reports. A number of network providers in the country have raised the issue of high pricing with the regulator and that discussion is currently still ongoing. The three player sector has a cumulative debt of nearly eight lakh crore, and many fear that the current pricing will just not allow the telcos to even bid for the 5G spectrum. However, all hope is not lost.

Reliance Jio already has an IP-ready network in place. The company has gone on record to say that they will be able to launch 5G telecom services within six months of spectrum allocation. Since the government has stated that it plans to allocate the airwaves for 5G by the end of 2019, Reliance Jio could be bringing 5G to the consumer market by mid-2020.

At MWC 2019, Bharti Airtel and Cisco announced a collaboration to build India's "largest 5G-ready IP network". This represents a reversal from the company's statements from last year where they said they won't participate in an upcoming 5G auction since they felt ultra fast wireles was still "three to four years away" in addition to the cash flow problems they've been having recently.

Network vendors like Huawei and Nokia are positive regarding the rollout of 5G in the country, and have only raised concerns over the availability of spectrum. If there is some way the Catch 22-like situation with the poorly performing telcos is resolved, 5G rollout in India by late 2020-early 2021 could very much be a possibility.